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Optimizing Portfolio Performance through Clustering and Sharpe Ratio-Based Optimization: A Comparative Backtesting Approach

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Optimizing portfolio performance is a fundamental challenge in financial modeling, requiring the integration of advanced clustering techniques and data-driven optimization strategies. This paper introduces a comparative backtesting approach that combines clustering-based portfolio segmentation and Sharpe ratio-based optimization to enhance investment decision-making. First, we segment a diverse set of financial assets into clusters based on their historical log-returns using K-Means clustering. This segmentation enables the grouping of assets with similar return characteristics, facilitating targeted portfolio construction. Next, for each cluster, we apply a Sharpe ratio-based

optimization model to derive optimal weights that maximize risk-adjusted returns. Unlike traditional mean-variance optimization, this approach directly incorporates the trade-off between returns and volatility, resulting in a more balanced allocation of resources within each cluster. The proposed framework is evaluated through a backtesting study using historical data spanning multiple asset classes. Optimized portfolios for each cluster are constructed and their cumulative returns are compared over time against a traditional equal-weighted benchmark portfolio.

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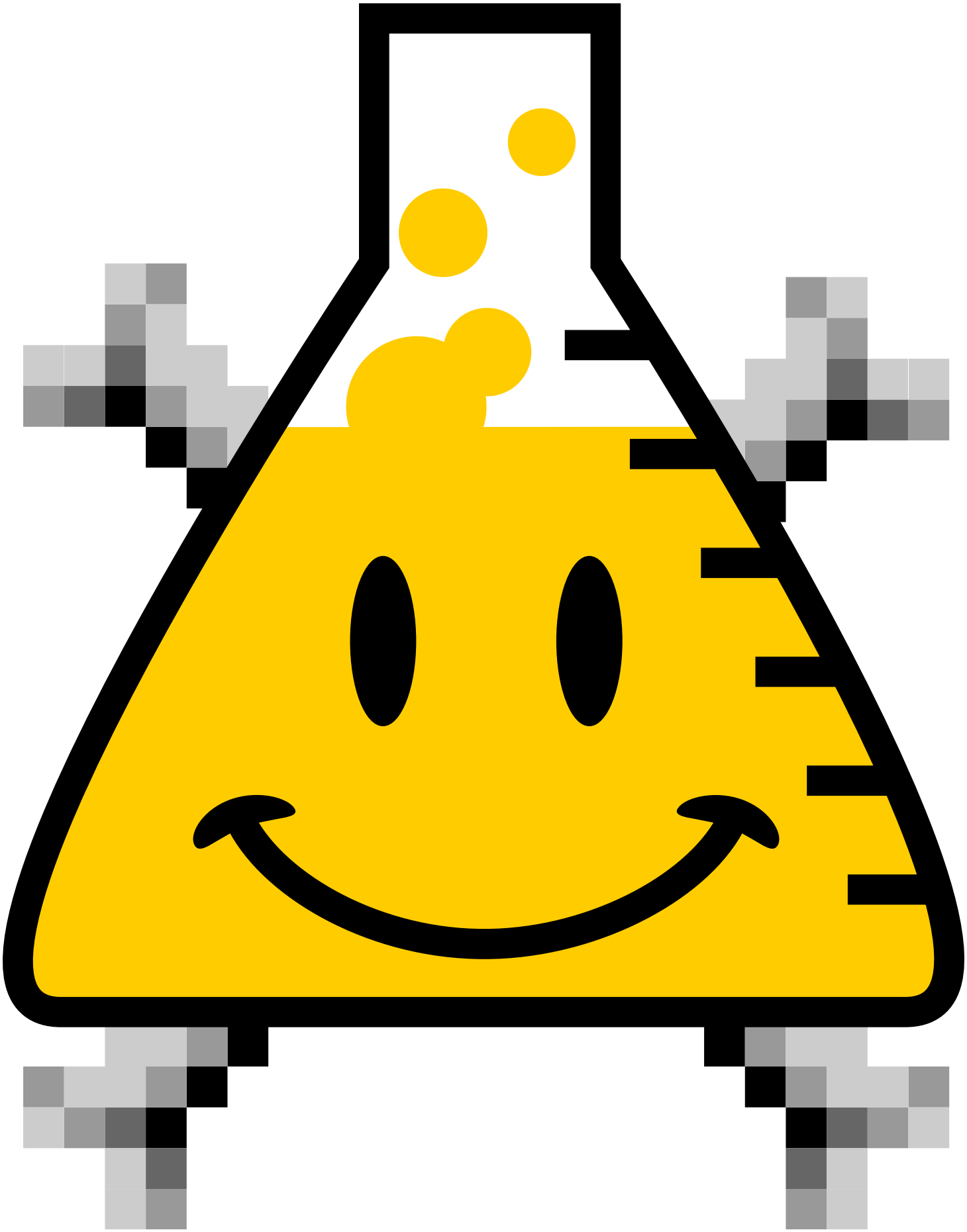

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